

AI IMAGE LAB: TECHNICAL SPECIFICATION

FakeShield AI Image Lab: Technical Specification & Forensic Architecture

1. Executive Overview

The **AI Image Lab** is the flagship image forensic module of the FakeShield ecosystem. It represents a 2026-standard implementation of multi-signal deepfake detection, specifically engineered to counter modern diffusion models (SDXL, Midjourney v7, DALL-E 3) and generative water-filling techniques. Unlike binary classifiers, the Lab utilizes a **Confidence-Weighted Fusion Engine** that evaluates evidence across geometric, semantic, spectral, and textural domains.

2. Theoretical Architecture: The Multi-Signal Suite

2.1 RIGID: Perturbation Invariance Analysis

- **Backbone:** `facebook/dinov2-base` (Vision Transformer with Self-Supervised Learning).
- **Theory:** Real-world photographs possess physical "rigidity." When subjected to Gaussian noise, their core structural embeddings remain invariant. AI-generated images, which exist on a highly sensitive latent manifold, exhibit "fragility"—their embeddings collapse or shift significantly under minimal perturbation.
- **Metric:** The system calculates the Cosine Similarity between the original image and 8 noisy variants. A similarity score < 0.85 triggers an AI flag.

2.2 Neural Ensemble: Spatial Texture Fingerprinting

- **Models:** Ensemble of `umm-maybe` (SigLIP-based) and `dima806` (ViT-based) detectors.
- **Theory:** Generative models leave "spatial fingerprints"—micro-patterns in the arrangement of pixels that differ from the Bayer filter patterns of physical camera sensors.
- **Fusion Logic:** The engine weighs SOTA research models to avoid false positives on high-frequency natural textures like sand, fabric, or complex depth-of-field blur.

2.3 CLIP Semantic Domain Gap

- **Strategy:** Zero-shot contrastive analysis using `openai/clip-vit-large-patch14`.
- **Methodology:** The image is projected into a shared latent space alongside carefully engineered prompt pairs.
- **Significance:** This signal catches AI "aesthetics" (e.g., the hyper-realistic smoothness of Midjourney or the lighting inconsistencies of DALL-E) that are chemically identified as "synthetic" by the CLIP transformer.

2.4 Spectral Forensics: FFT Power Decay

- **Math:** 2D Fast Fourier Transform (FFT) + Radial Integral Operation (RIO).
- **Discovery:** Natural photography follows a power-law decay ($1/f^2$). GANs and Diffusion models introduce high-frequency "plateaus" due to upsampling artifacts and grid-based generation.
- **Diagnostic:** A deviation in the spectral slope (α) from the natural -2.3 characteristic is a definitive forensic marker.

2.5 PRNU & Noise Residue

- **Basis:** Noiseprint + Photo Response Non-Uniformity (PRNU) proxies.
- **Process:** Median filter residuals isolate the sensor noise. Real cameras show structured noise (fixed-pattern noise); AI images show isotropic (uniform) noise.
- **Metrics:** Analysis of Noise Kurtosis and Isotropy.

3. The "Hard Veto" Short-Circuit Pipeline

FakeShield employs a tiered verification system to ensure maximum recall while maintaining zero-latency for known signatures.

Tier 1: C2PA Content Credentials

- **Protocol:** CAI (Content Authenticity Initiative).
- **Action:** If a cryptographic manifest is present (e.g., from DALL-E 3 or Firefly) declaring the image as "GenAI," the analysis exits immediately with a **Critical AI Verdict**.

Tier 2: Gemini Watermark Engine

- **Target:** Google Gemini (Imagen) 4-pointed sparkle.
- **Algorithm:** Dual-stage template matching combined with geometric verification.
- **Veto:** Includes a "Saturation Veto" (to ignore bright fabric) and "Point Check" (ensuring the tips of the astroid are distinct).

Tier 3: Tampering/Removal Detection

- **Concept:** "The Absence of Evidence is Evidence."
- **Logic:** Detecting "Heal" or "Inpaint" anomalies in the bottom-right corner where watermarks are typically located. If a user deletes a watermark, FakeShield identifies the local variance delta and flags the image as AI-generated.

4. Forensic Workflow & Fusion Engine

4.1 Pre-processing

- Standardization to RGB space.
- Application of Hann windows for spectral stability.
- High-pass filtering for noise isolation.

4.2 Parallel Inference

All non-veto signals (RIGID, Neural, CLIP, FFT, Noise, ELA) run in a `ThreadPoolExecutor`. This reduces total forensic latency to < 2.5 seconds for high-resolution assets.

4.3 Fusion Logic (v2026 Calibration)

The final probability is calculated as:

$$P_{\text{final}} = \frac{\sum (\text{Score}_i \times \text{Power}_i)}{\sum \text{Power}_i}$$

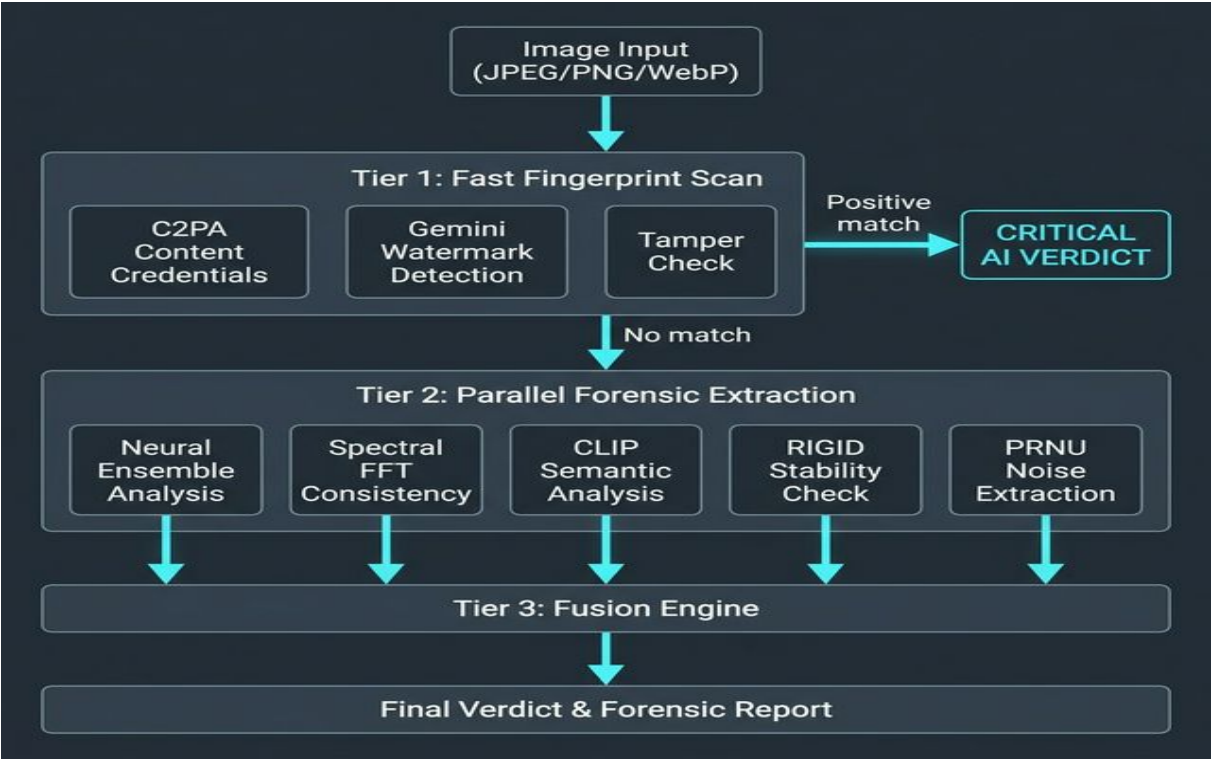
Where Power_i is a dynamic factor derived from the signal's weight and real-time confidence (e.g., a low-quality JPEG downweights the FFT signal).

5. Visual Forensic Capabilities

The AI Image Lab goes beyond a simple probability, providing tools for human-in-the-loop verification:

- **Heatmap Overlay:** JET-mapped noise residuals highlighting manipulation zones.
- **ELA Viewer:** Error Level Analysis emphasizing re-compression deltas.
- **Spectrum Map:** Magma-mapped 2D FFT highlighting upsampling artifacts.
- **Forensic Lens:** Depth-of-field and metadata inspector.

6. Logic Flowchart: Adaptive Forensic Pipeline



NOTE: FakeShield AI Image Lab is designed for forensic-grade applications. It provides a detailed traceability report for every verdict, ensuring that automated scores are backed by clinical evidence.